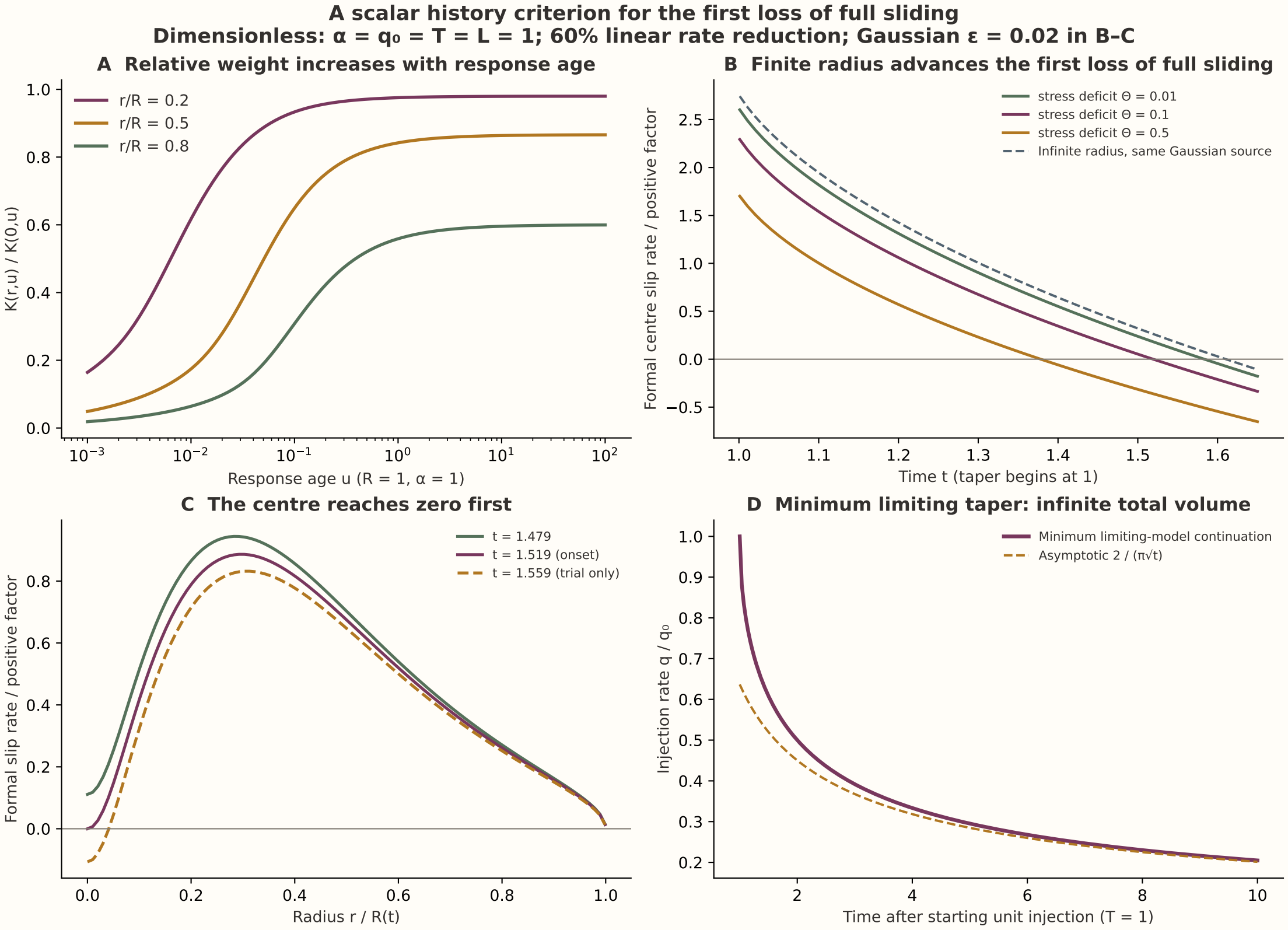
A scalar history criterion for the first loss of full sliding
Dimensionless: α = q₀ = T = L = 1; 60% linear rate reduction; Gaussian ε = 0.02 in B–C
A Relative weight increases with response age
r/R = 0.2
r/R = 0.5
r/R = 0.8
K(r,u) / K(0,u)
Response age u (R = 1, α = 1)
B Finite radius advances the first loss of full sliding
stress deficit Θ = 0.01
stress deficit Θ = 0.1
stress deficit Θ = 0.5
Infinite radius, same Gaussian source
Formal centre slip rate / positive factor
Time t (taper begins at 1)
C The centre reaches zero first
t = 1.479
t = 1.519 (onset)
t = 1.559 (trial only)
Formal slip rate / positive factor
Radius r / R(t)
D Minimum limiting taper: infinite total volume
Minimum limiting-model continuation
Asymptotic 2 / (π√t)
Injection rate q / q₀
Time after starting unit injection (T = 1)